WORKED SOLUTION

STEP I 2008, Question 5

Pure • Polynomial Bounds

STEP I 2008, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s1-q5

The key idea

For $n = 1$ the polynomial is a straight line, and its values at the two ends of the interval are tied together: they differ by exactly 2, so they can't both be small. In part (ii) the trick is to divide by the leading coefficient so that Chebyshev's theorem applies to the curve itself. The theorem then forces $|y|$ to be large somewhere in $[-1, 1]$, while the values at the endpoints are small. A curve that climbs higher than both of its endpoint values (or dips lower than both) must turn round somewhere in between.

Solution

Part (i)

The case $n = 1$. Here $\mathrm{p}(x) = x + a_0$, and we need $M \geq 2^0 = 1$. Since M is at least the value of $|\mathrm{p}|$ at any point of $[-1, 1]$, it's at least the values at the endpoints:

$$M \geq |\mathrm{p}(1)| = |1 + a_0| \quad \text{and} \quad M \geq |\mathrm{p}(-1)| = |a_0 - 1|.$$

Adding, and using the triangle inequality $|u| + |v| \geq |u - v|$,

$$2M \geq |1 + a_0| + |a_0 - 1| \geq |(1 + a_0) - (a_0 - 1)| = 2,$$

so $M \geq 1 = 2^{1-n}$, which proves the theorem for $n = 1$. (Equality holds for $\mathrm{p}(x) = x$, so the bound can't be improved.)

(a) For $\mathrm{p}(x) = x^2 - \frac{1}{2}$ and $|x| \leq 1$ we have $0 \leq x^2 \leq 1$, so $-\frac{1}{2} \leq \mathrm{p}(x) \leq \frac{1}{2}$. The value $\frac{1}{2}$ is attained at $x = \pm 1$ (and $-\frac{1}{2}$ at $x = 0$), so

$$M = \tfrac{1}{2} = 2^{1-2}.$$

The theorem holds, with equality.

(b) For $\mathrm{p}(x) = x^3 - x$ we need $M \geq 2^{1-3} = \frac{1}{4}$. It's enough to find a single point where $|\mathrm{p}| \geq \frac{1}{4}$. The stationary points are where $3x^2 - 1 = 0$, that is $x = \pm\frac{1}{\sqrt{3}}$, both in $[-1, 1]$,

and

$$\mathrm{p}\left(-\tfrac{1}{\sqrt{3}}\right) = -\frac{1}{3\sqrt{3}} + \frac{1}{\sqrt{3}} = \frac{2}{3\sqrt{3}}.$$

Since $\mathrm{p}(\pm 1) = 0$, this is in fact the greatest value: $M = \frac{2}{3\sqrt{3}} = \frac{2\sqrt{3}}{9} \approx 0.385$. To compare with $\frac{1}{4}$ without a calculator, square both sides:

$$\left(\frac{2}{3\sqrt{3}}\right)^2 = \frac{4}{27} > \frac{4}{64} = \frac{1}{16} = \left(\frac{1}{4}\right)^2,$$

so $M > \frac{1}{4}$ and the theorem holds (this time with room to spare).

Part (ii)

Write $f(x) = 64x^5 + 25x^4 - 66x^3 - 24x^2 + 3x + 1$. Chebyshev's theorem is about polynomials with leading coefficient 1, so we apply it to

$$\mathrm{p}(x) = \tfrac{1}{64} f(x) = x^5 + \tfrac{25}{64}x^4 - \tfrac{66}{64}x^3 - \tfrac{24}{64}x^2 + \tfrac{3}{64}x + \tfrac{1}{64},$$

which has the required form with $n = 5$. The theorem gives $\max_{|x|\le 1} |\mathrm{p}(x)| \ge 2^{-4} = \frac{1}{16}$, and multiplying by 64,

$$\max_{|x|\le 1} |f(x)| \ge 4.$$

So there is some $x_0 \in [-1, 1]$ with $|f(x_0)| \ge 4$.

Now look at the endpoints:

$$f(1) = 64 + 25 - 66 - 24 + 3 + 1 = 3,$$
$$f(-1) = -64 + 25 + 66 - 24 - 3 + 1 = 1.$$

Both have modulus less than 4, so x_0 is not an endpoint: $-1 < x_0 < 1$.

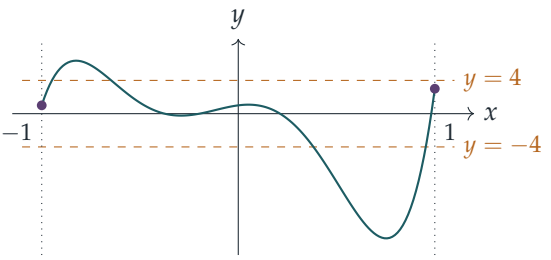
Ask yourself We know $|f|$ gets big inside the interval. Why does that force a turning point?

There are two cases.

- If $f(x_0) \ge 4$, then $f(x_0)$ is bigger than both $f(-1) = 1$ and $f(1) = 3$. The continuous function f attains a greatest value on the closed interval $[-1, 1]$; this value is at least $f(x_0) \ge 4$, so it isn't attained at either endpoint. It is therefore attained at an interior point, which is a local maximum of f.
- If $f(x_0) \le -4$, then $f(x_0)$ is below both endpoint values, and in the same way the least value of f on $[-1, 1]$ is attained at an interior point, which is a local minimum.

Either way f has a local maximum or minimum strictly inside $(-1, 1)$. There $f' = 0$, and since f is a non-constant polynomial, f' has only finitely many zeros. So f' keeps one sign just to the left of the point and one sign just to the right. Because the point is a local maximum or minimum, those signs must be opposite, so it is a genuine turning point. Hence the curve has at least one turning point with $-1 \le x \le 1$, as required.

Notice that we never had to solve the quartic $f'(x) = 0$. That's the point of the question.



The sketch shows the curve on $[-1, 1]$. Both endpoint values lie between the dashed lines $y = \pm 4$, but the curve goes outside that band. In fact it has four turning points in the interval, the deepest near $x \approx 0.75$ with $y \approx -15$.

Checks

- For $n = 1$, trying a few values of a_0 shows that $\max(|1 + a_0|, |1 - a_0|)$ is never below 1, and equals 1 only when $a_0 = 0$.
- In (b), $\frac{2}{3\sqrt{3}} \approx 0.3849$, comfortably above 0.25. Sampling $x^3 - x$ on a fine grid confirms this is the maximum of $|x^3 - x|$.
- In (ii), a numerical scan of f on $[-1, 1]$ gives $\max |f| \approx 15.0$ (at $x \approx 0.753$, where $f \approx -15.0$), well above the guaranteed 4. It also finds f' changing sign at $x \approx -0.825$, -0.292, 0.052, 0.753.

Common slips

- Applying the theorem to f directly. Its leading coefficient is 64, not 1, so you must divide by 64 first. The bound becomes $64 \cdot 2^{-4} = 4$.
- Stopping at "$|f| \geq 4$ somewhere". That alone gives no turning point: you need the endpoint values 1 and 3 to show that the extreme value is reached inside the interval.
- In (b), comparing $\frac{2}{3\sqrt{3}}$ with $\frac{1}{4}$ carelessly. Squaring both positive numbers makes the comparison exact.

Beyond STEP

Equality in Chebyshev's theorem holds exactly for $2^{1-n} T_n(x)$, where T_n is the Chebyshev polynomial defined by $T_n(\cos\theta) = \cos n\theta$. Part (a) is $\frac{1}{2}T_2$, and the $n = 1$ case is $T_1(x) = x$. These polynomials are the starting point of approximation theory: they explain why interpolation works best at Chebyshev nodes rather than at equally spaced points. The argument in part (ii) uses the extreme value theorem (a continuous function on a closed interval attains its bounds), which is proved carefully in a first analysis course. Abbott's *Understanding Analysis* is a good place to read it.

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